

PROJECT BOATY

Subsystem Specification Simulation & test

Let the whole system be exercised, failed on purpose and rehearsed at home, with the same Mission Control software as on the bank.

Document	BOATY-SSS-SIM
Issue	Issue A (for review)
Date	28 September 2026
Status	For review by the project owner
Parents	SRS Issue D, ADD Issue C, ICD Issue B
Content	3 allocated SRS requirements → 12 subsystem requirements (9 M, 3 S, 0 C)

Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review.	Claude (drafted)

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: section 5 is what will be built and tested. Check that each requirement is right, sufficient and testable. Section 4 shows every parent requirement is covered (the build script checks this).

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1. Scope

1.1 Purpose

Let the whole system be exercised, failed on purpose and rehearsed at home, with the same Mission Control software as on the bank.

1.2 Boundary

Inside this subsystem	Outside (and who owns it)
• ArduPilot SITL (Rover, boat frame) • Camera stub (IF-03) • Launch scripts • Scenario runner and test suites • CI configuration	• The software under test (MCN, MCP services) • Hardware tests (bench rigs are per-subsystem)

1.3 Breakdown

Item	Description
SIM-1 Simulator	SITL + camera stub + launcher
SIM-2 Test suites	Unit, contract, scenario, LLM evaluation
SIM-3 CI	GitHub Actions workflows

2. Applicable documents

Document	Use
BOATY-SRS-001 Issue D	Parent requirements
BOATY-ADD-001 Issue C	Allocation, decisions, budgets, early verification items
BOATY-ICD-001 Issue B	Interface definitions (all listed in section 3 apply in full)

3. Interfaces

The subsystem shall meet every interface below as defined in the ICD. 'Owner' means this subsystem maintains the definition.

Interface	Other end	Role	ICD
IF-15 Site data	Mission Control	Conforms	\$5
IF-21 Simulation link	Mission Control	Owner	\$5

4. Parent requirements

Primary (3): this subsystem is responsible for meeting and verifying these SRS requirements.

SRS	Pri	Requirement (abridged)	Met by
SWE-004	M	The whole system shall run in simulation (helm simulator and simulated camera) on the Pi 5 or a laptop, with Mission...	SIM-D01, SIM-D02, SIM-D03, SIM-D04
SWE-005	M	Each failsafe requirement (FS-001 to FS-013) shall have an automated simulation scenario.	SIM-D06, SIM-D07
SWE-007	S	Unit tests (pytest), linting (ruff) and type checking (mypy) shall run automatically on every push.	SIM-D11

5. Subsystem requirements

Requirements are numbered SIM-Dnn. Pri, Ver and Stage follow SRS conventions (M/S/C; Test, Demonstration, Inspection, Analysis, Simulation; SIM, BENCH, POOL, LAKE). 'Trace' lists the parent SRS requirements, interfaces (IF), decisions (DD) and early verification items (V) it implements.

5.1 Simulator

ID	Requirement	Pri	Ver	Stage	Trace
SIM-D01	SITL shall run the same ArduPilot version as the helm, with a skid-steer boat frame and the controlled parameter file.	M	D	SIM	SWE-004, V-12
SIM-D02	The camera stub shall implement IF-03 exactly, serving test images geotagged from SITL.	M	T	SIM	SWE-004, IF-21
SIM-D03	One command shall start the simulator on a laptop or the Pi 5 at a given site's home.	M	D	SIM	SWE-004
SIM-D04	Mission Control shall run against the simulator with configuration changes only.	M	D	SIM	SWE-004, SWE-003
SIM-D05	A rehearsal mode shall run the physical panel against the simulator for crew practice.	S	D	SIM	OPS-012

5.2 Tests

ID	Requirement	Pri	Ver	Stage	Trace
SIM-D06	Each FS-001 to FS-013 requirement shall have an automated scenario with explicit pass criteria (section 6). FS-008 is bench-only and is marked as such.	M	T	SIM	SWE-005
SIM-D07	Fault injection shall use SITL simulation parameters and network impairment, mapped per scenario.	M	T	SIM	SWE-005, IF-21
SIM-D08	Contract tests for IF-03 and IF-14 shall run against both the simulator and real hardware.	M	T	SIM	IF-14, IF-03
SIM-D09	The validator's adversarial suite (≥ 50 cases) shall run in CI.	M	T	SIM	VAL-006
SIM-D10	The LLM evaluation set (≥ 30 instructions) shall run on demand with a per-run cost cap.	S	T	SIM	NLI-003, NLI-006

5.3 CI and reproducibility

ID	Requirement	Pri	Ver	Stage	Trace
SIM-D11	Every push shall run ruff, mypy and unit tests. SITL scenarios shall run on pull requests and nightly.	S	I	SIM	SWE-007
SIM-D12	Tool and dependency versions (SITL commit, Python lock file) shall be pinned in the repository.	M	I	SIM	SWE-006

6. Failsafe scenario catalogue

Req	Scenario	Injection	Pass criterion
FS-001	Battery drain to 35% then 15%	Simulated battery capacity/drain	RTL at 35%; alarm at 15%; reaches home
FS-002	Drop UDP 14550 in STEERING	Network impairment	HOLD $\leq$ 2 s; RTL at 10 s (B4)
FS-003	Drop UDP 14550 in AUTO	Network impairment	Mission continues; RTL at 60 s (B4)
FS-004	GNSS failure for 5 s then restore	SITL GPS failure parameter	Motors stop $\leq$ 3 s; RTL after 10 s healthy (B6)
FS-005	Heavy added drag, throttle $\geq$ 50%	SITL drag/wind parameters	HOLD $\leq$ 5 s + event
FS-006	As FS-005, drag released after burst 2	SITL + B5	$\leq$ 3 bursts; resumes mission
FS-007	Kill all MCP services mid-mission	Stop the camera stub and B-services	Mission completes; RTL; HOLD at home
FS-008	Helm signal loss to ESCs	Bench only	Motors stop $\leq$ 1 s (PRP-D07)
FS-009	Press STOP (panel or API) in each state	Panel emulation	Motors stop $\leq$ 1 s
FS-010	Moisture flag set in the stub	Stub health = moisture	RTL $\leq$ 2 s
FS-011	GNSS loss during battery RTL	Combined injections	Motors stop (HOLD) wins
FS-012	All of the above	Log inspection	Every event logged and announced $\leq$ 2 s
FS-013	Sweep of single faults across mission phases	Scripted matrix	Never leaves the inclusion fence under power

Injection parameter names are fixed per SITL version (TBC-04) and kept in the scenario code, not here.

7. Design constraints and budgets

- Same parameter file as the real helm, plus SIM_* overrides
- Runs on a laptop or the Pi 5

Budget	Allocation
Cost	£0 (open-source tools); LLM evaluation runs cost API credit, so they're capped
Mass / power	n/a

8. Verification and open items

Each requirement is verified at its stage using the method shown. Test procedures are written per stage and reference these IDs.

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	6	4	2	-	-	12
BENCH	-	-	-	-	-	0
POOL	-	-	-	-	-	0
LAKE	-	-	-	-	-	0

Ref	Open item
TBC-03 / V-12	SITL skid boat frame
TBC-04	Fault injection parameters per scenario